

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: COMPUTER VISION with OpenCV

COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)*

Assessment Tool Weightage: 50%

1. Grey Level Transformation

(a) Purpose

Grey level transformation changes pixel intensity values to improve image brightness and contrast.

(b) Negative Transformation

Formula:

$$s = 255 - r$$

r	s = 255 - r
50	205
100	155
150	105
200	55

Answer: [205, 155, 105, 55]

2. Log Transformation

(a) Purpose

Log transformation enhances dark regions by expanding low-intensity values and compressing high-intensity values.

(b) Given

$$s = 10\log(1 + r)$$

(using log base 10)

r	s
0	0
10	10.41
50	17.08
100	20.04

Answer: [0, 10.41, 17.08, 20.04]

3. Histogram Processing

(a) Purpose

Histogram equalization improves image contrast by redistributing pixel intensities.

(b)

Histogram

Intensity	Frequency
0	2
1	2
2	4
3	8

Total pixels = 16

CDF

Intensity	CDF
0	2
1	4
2	8
3	16

Equalized value

$$s = \frac{CDF}{16} \times (4 - 1)$$

Intensity	New Value
0	0
1	1
2	2
3	3

4. Spatial Filtering (Smoothing)

(a) Purpose

Smoothing removes noise and small variations by averaging neighboring pixels.

(b)

Matrix

10 20 30

20 30 40

30 40 50

Average

$$= \frac{10 + 20 + 30 + 20 + 30 + 40 + 30 + 40 + 50}{9} = \frac{270}{9} = 30$$

Answer: 30

5. Spatial Filtering (Sharpening)

(a) Purpose

Sharpening highlights edges and fine image details.

(b)

Mask

0 -1 0

-1 4 -1

0 -1 0

Matrix

10 10 10

10 20 10

10 10 10

Output

$$= (4 \times 20) - (10 + 10 + 10 + 10) = 80 - 40 = 40$$

Answer: 40

6. Frequency Domain Filtering

(a) Difference

Low-pass filter: Removes high frequencies and smooths the image.

High-pass filter: Removes low frequencies and enhances edges.

(b)

Given

$$F = [100, 80, 40, 20]$$

Retain only first two components.

Filtered Output

$$[100, 80, 0, 0]$$

7. Thresholding

(a) Purpose

Global thresholding separates foreground and background using one threshold value.

(b)

Pixels

20 80 120 200

Threshold

$$T = 100$$

Rule

- Pixel $\geq 100 \rightarrow 1$
- Pixel $< 100 \rightarrow 0$

Output

0 0 1 1

8. Edge Detection

(a) Purpose

Edge detection identifies object boundaries where intensity changes rapidly.

(b)

Horizontal Sobel

-1 -2 -1

0 0 0

1 2 1

Image

10 10 10

20 20 20

30 30 30

Calculation

$$= (-10 - 20 - 10) + (30 + 60 + 30) = 80$$

Gradient = 80

9. Power-Law (Gamma) Transformation

(a) Purpose

Gamma correction adjusts image brightness.

(b)

Formula

$$s = r^{0.5}$$

(c) & (d)

r	Output
25	5
100	10
225	15

Answer: [5,10,15]

(e)

Brightness of darker pixels increases, making the image appear brighter.

10. Gaussian Smoothing

(a) Purpose

Reduces noise while preserving edges better than averaging.

(b)

Gaussian filter gives more weight to the center pixel.

(c)

Kernel

1 2 1

2 4 2

1 2 1

Image

10 20 10

20 40 20

10 20 10

Weighted sum

$$\begin{aligned} &= \frac{(1 \times 10) + (2 \times 20) + (1 \times 10) + (2 \times 20) + (4 \times 40) + (2 \times 20) + (1 \times 10) + (2 \times 20) + (1 \times 10)}{16} \\ &= \frac{360}{16} = 22.5 \end{aligned}$$

Center Pixel = 22.5 $\approx$ 23

(e)

Noise is reduced while preserving image details.

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method

Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation